Professional Certificate of Competency in Electrical Power System Fundamentals for Non-Electrical Engineers

Webinar 8

Module 9: Impact of Renewables on Grid and Power Quality

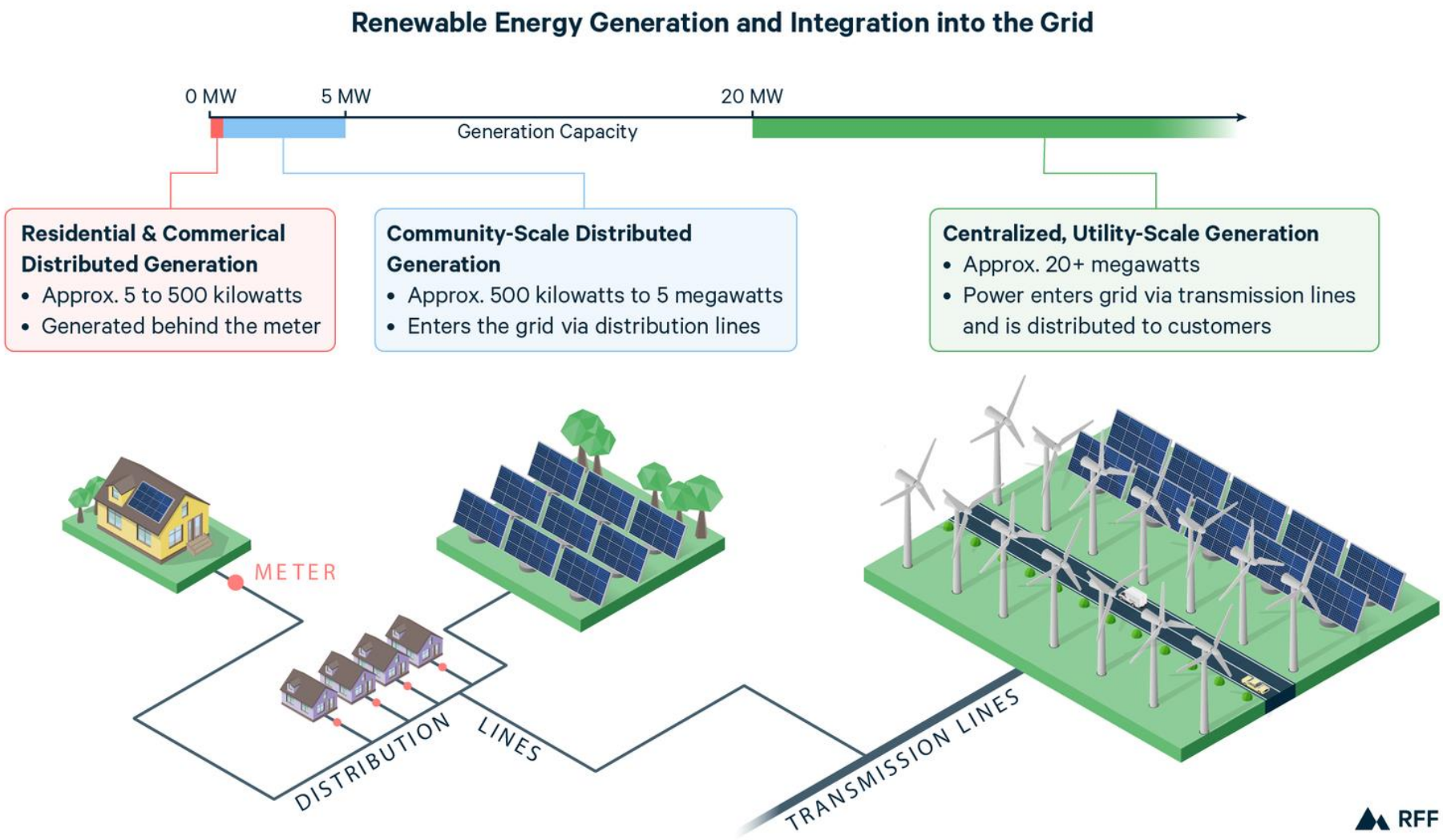
Module 9 Topics

- Impact of renewables on grid
- Power quality problems: voltage and frequency variations, interruptions, lightning, surges and harmonics
- Power quality improvement and mitigation

Renewable Energy Integration

- The **goal** of renewable energy integration is to advance system design, planning and operation of the electric grid to:
  - reduce carbon emissions and emissions of other air pollutants through increased use of renewable energy and clean distributed generation
  - increase asset use through integration of distributed systems and customer loads to reduce peak load and lower the costs of electricity
  - support achievement of renewable portfolio standards for renewable energy and energy efficiency
  - enhance reliability, security, and resiliency from microgrid applications in critical infrastructure protection and constrained areas of the grid
  - support reductions in oil use by enabling plug-in electric vehicle (PHEV) operations with the grid

Renewable Energy Integration



Renewable Energy Integration

Utility-scale generation:

- Centralized, utility-scale renewable energy plants are comparable to fossil-fueled power plants and can generate between several and hundreds of megawatts (MW) of power. This is sent across transmission lines, converted to lower voltage, and transmitted across distribution lines to buildings and home.
- Unlike conventional, fossil-fuel plants, however, renewable energy plants are typically not **dispatch able** (or able to generate power when called upon), because they depend on variable resources like the sun and wind that change over the course of a day.
- Since wind and solar power have zero fuel costs, they get first priority in the dispatch order, meaning that they are used before other generator types.

Renewable Energy Integration

Distributed generation:

- Small residential and commercial renewables typically range between 5 and 500 kilowatts (kW). Most of these small renewables are solar panels, which are easily customizable in size.
- These distributed resources are typically located on-site at homes or businesses (such as rooftop solar panels).
- Unlike large, centralised renewable plants that connect to the grid through high-voltage transmission lines, distributed resources like these are connected to the grid through low voltage distribution network that delivers electricity to customers.

Renewable Energy Integration

Distributed generation:

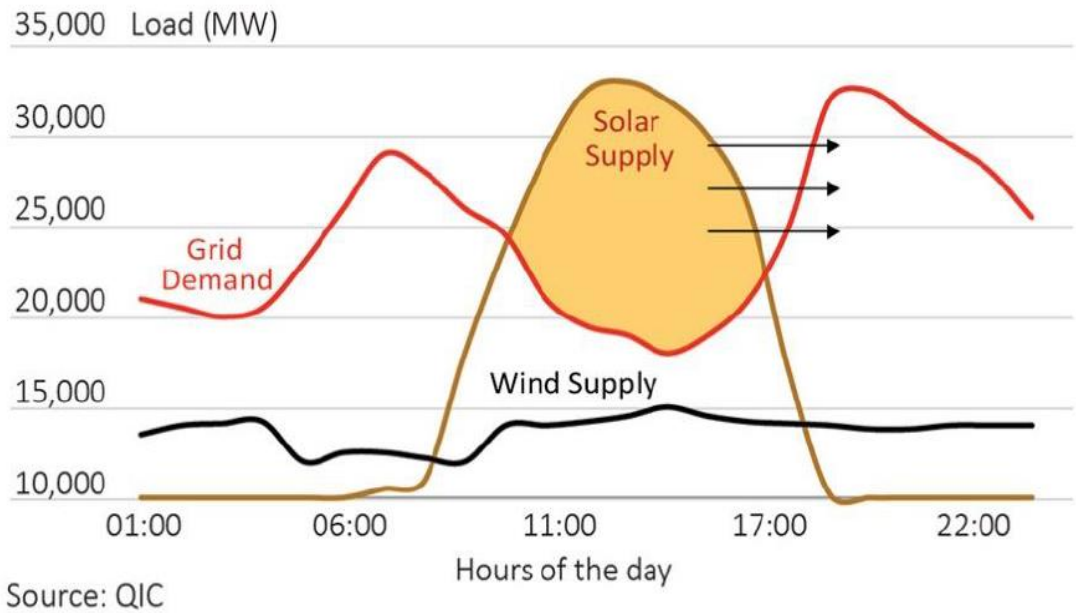
- Community-scale renewables, which are larger than rooftop projects but smaller than utility-scale, are also connected to the grid through distribution lines and therefore also distributed generation.
- Unlike small rooftop renewables, however, community-scale renewables reside "**in front of the meter**," meaning that the power they generate is not used on-site (e.g.: home) but rather flows onto the distribution grid to be used by homes and businesses in the near vicinity.

Renewable Energy Integration

- Most of the time, the grid will absorb all of the electricity produced by renewables because there is sufficient demand for electricity.
- Consequently, grid operators only need to use other sources to make up the difference between the amount of electricity demanded and the amount of electricity produced by renewables on the grid. This is known as **net load**.
- Utilities are responsible for meeting the net load and typically use conventional fossil-fuel resources, like natural gas plants, to do so.
- As a result, the more renewable energy resources present on the grid, the less electricity must be generated using conventional fossil-fuel plants.

Intermittent Nature of Renewables

- Fluctuating resources
- Variation over the day
- Difficulty in matching demand with supply
- Need additional supply security measures, e.g. energy storage
- Increases the cost of system



http://www.qic.com.au/knowledge-centre/technology-disruptions-affecting-infrastructure-20160414

Renewable Energy - Grid Constraints

- Wind and solar energy are an intermittent source of power.
- Since weather can change quickly and unpredictably, high renewables penetration requires grid operators to quickly react to new production patterns. Failure to do so could lead to power shortages and blackouts.
- Even if the weather is predictable, grid operators face the issue of how to quickly respond to dwindling production from solar energy when the sun goes down but the demand for electricity stays the same or increases.
- Conventional power plants or storage facilities must deal with these variations.
- This means that output variations in wind or solar power can give rise to extra cost and complexity due to start/stop of conventional units.
- Other issues include voltage and frequency regulation, equipment rating etc.

Voltage Regulation

Solutions to overcome undesirable effect of large Photo Voltaic (PV) solar export:

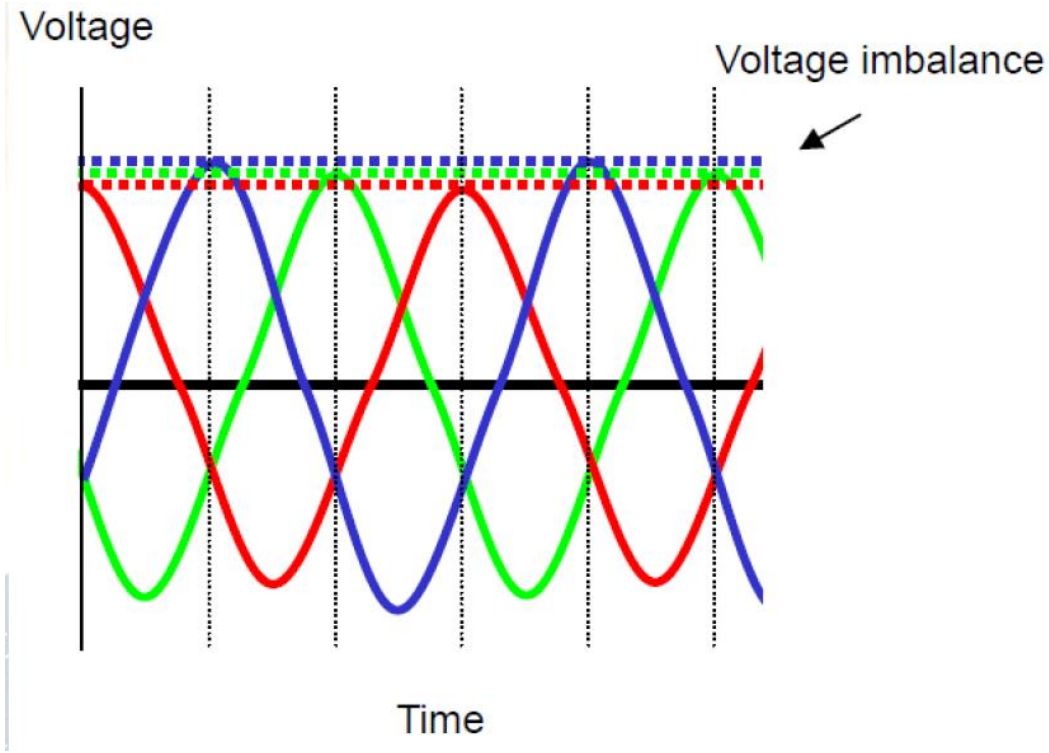
- PV curtailment (inverters switches off when grid voltage is a certain value)
- Installation of voltage regulating devices
- Transformer on load tap adjustment (voltage control)
- PV converter reactive power compensation (also controls power factor)
- Systems designed to regulate grid voltage within the allowable limit by consuming surplus energy or regulating reactive power

Frequency Regulation

- The power system inertia decreases with increased share of renewables.
- Hence, renewable power plants must adjust their output power proportionally to respond to frequency deviations.
- The PV controller functionality can be used to provide frequency control.

Voltage Imbalance

Voltage imbalance is a condition in which the amplitude of each phase voltage is different in a three-phase system or the phase difference is not exactly 120°.



Source: http://www.iea-pvps-task10.org/IMG/pdf/rep10_06.pdf

Harmonics

The **harmonic of a wave** is defined as a component frequency of the signal that is an integer multiple of the fundamental frequency.

- Grid load such as from appliances and computers use power electronics technologies to change the grid AC to the desired current waveform.
- In this process, these devices generate “harmonics” that may distort the grid waveform.

Connection Rating Checks

- The existing grid infrastructure may be serviceable for existing loads.
- New major generation connections can result in a higher energy flow which may stress the weaker spots of the grid (e.g.: conductor and cable joints, tap changer rating for older transformers).
- As part of the connection agreement, the network may need to inspect, test and review their asset ratings before approving the connection.
- Any costs for rectification or equipment upgrade to service the new connection and ongoing maintenance costs are passed to the connection proponent.
- There may also be a need for voltage regulating devices and those costs may also be considered for pass through.

Renewable Energy – Improve Resilience

There are several ways to **increase grid flexibility** and **improve the integration of renewable resources:**

- **Energy storage** can be paired with variable renewables to accommodate fluctuations in renewable generation over the course of the day or several days. Electricity can be stored during times of high generation (for solar, during sunny days; for wind, during times of high wind speeds) for later use during periods of high demand.
- **Building more transmission lines** to connect areas endowed with plentiful renewable resources (e.g., very sunny or windy areas) with areas of high electricity demand can increase the value of renewable resources and reduce uncertainties of their electricity generation.

Renewable Energy – Improve Resilience

- **Combining different renewable sources** can balance out variability of production of one resource type. For example, a grid that relies on many types of renewables, such as solar, wind, hydro, and geothermal all together, will likely face less volatility in production than one that relies exclusively on wind energy alone.
- **Demand-side management** can include shifting flexible demand to times of high renewables production, such as preheating electric hot water heaters or charging electric cars during times of high solar energy production, and reducing demand during peak hours.

Renewable Energy – Improve Resilience

- Ensuring the network connection agreements outline stringent requirements including the need for gifted assets to comply with the network standards
- Ensuring an upfront inspection and upgrade of assets as needed (replace weak joints and connections) before connecting the new capacity
- Connection agreements to cover cost the network will incur due to capacity upgrades, network deterioration and additional inspections and maintenance which is recovered from the connection proponent
- Working closely with the Australian Energy Regulator to ensure appropriate rules changes to National Electricity Rules (NER) that allow the networks to be more innovative, flexible and agile to meet customer needs while maintaining network performance and lower cost of service. Standalone Power Supplies (SAPS) is an example.

Ancillary Services

- With rapidly increasing numbers of PV system installations, PV inverters are today also required to play a more active role in the grid system control.
- **Additional capabilities** are associated with:
  - Stabilising and controlling grid voltages
  - Reactive power control
  - Active filtering of harmonics
  - Balancing three-phase systems
  - Fault-ride through during voltage dips (through energy storage)
  - Shut down or reduction of output in case of over-production (requires remote control/communication means)

Power Quality

Power is considered to be of the required quality when made available:

- at stipulated voltage and frequency
- without distortion of waveform or loss of symmetry
- with minimum instances / duration of variations beyond the specified limits or unscheduled interruptions

Power Quality Attributes

Main attributes:

- Voltage within specified limits
- Frequency within specified limits

Others:

- Negligible instances of unscheduled power interruptions
- Control of transients or surges in the supply voltage
- In the case of a three phase supply, symmetry of voltages
- Sinusoidal waveform without appreciable distortion

Equipment Sensitivity is the Key

- General industrial equipment tolerance in the range of –10 to + 6% for sustained voltage variations and – 3 to +3 for frequency variations.
- Tolerance vs. time defined using different characteristic curves.

Voltage Variations

There could be different types of variations to the standard voltage of a system:

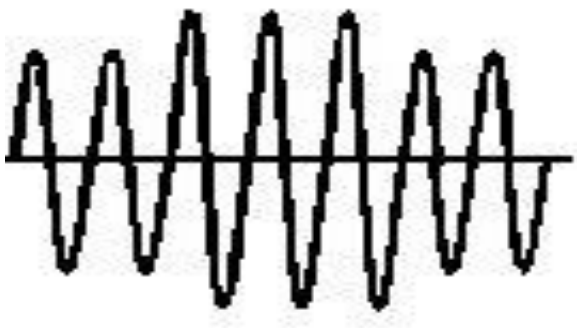
- Short duration:
  - Sag
  - Swell or surges
- Long duration:
  - Interruptions
  - Undervoltage
  - Overvoltage
- Voltage imbalance
- Voltage fluctuations

Short Duration Sags

- Decrease between 0.1 and 0.9 p.u. in rms voltage or current at the power frequency for duration from 0.5 cycles to 1 min.
- **Causes:**
  - Mostly local and remote faults
- Impact dropouts of sensitive customer equipment such as:
  - Computer crashes
  - Bulbs glow dim
  - Fan speed reduces
  - Effect on motor speed
  - Poor video quality of televisions, etc.

Short Duration Swells or Surges

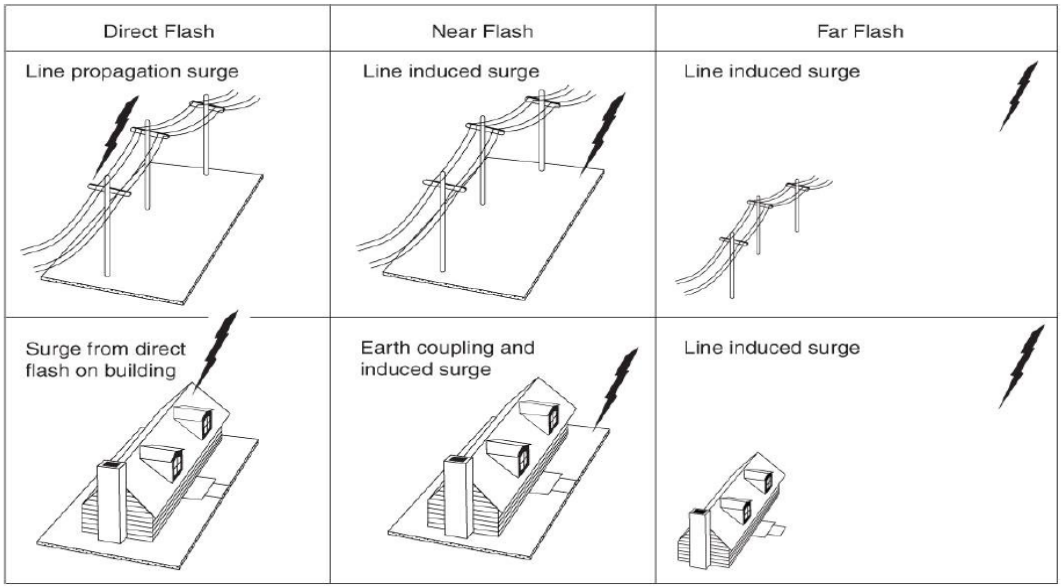
- Increase to between 1.1 and 1.8 p.u in the rms voltage or current at the power frequency for durations from 0.5 cycle to 1 min. Also called **spikes** or **transients.**
- **Causes:**
  - Single-line-to-ground faults.
  - Equipment switching over voltage.
  - Lightning strike



- **Impacts:**
  - Insulation stress, overheating of loads
  - Electronic equipment such as television, computers maloperation
  - Small fuses in electronic equipment blow off

Lightning Surges

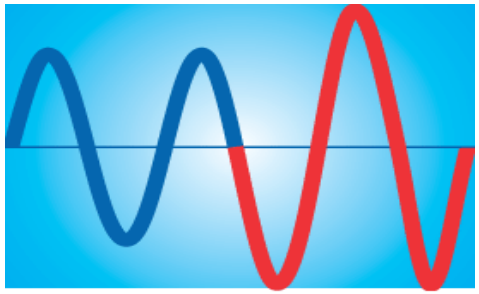
- Lightning surges in electrical systems can generally be classified according to their origin as follows:
  - Direct flashes to overhead lines
  - Induced over voltages on overhead lines
  - Over voltages caused by coupling from other systems.



https://leonardo-energy.pl/wp-content/uploads/2017/03/EIM-Przepi%C4%99cia-i-pr%C4%85dy-przej%C5%9Bciowe.pdf

Over Voltages

- Increase in the rms ac voltage greater than 110% at the power frequency for a duration longer than 1 min.
- **Causes:**
  - Load switching off
  - Capacitor switching on
  - System voltage regulation

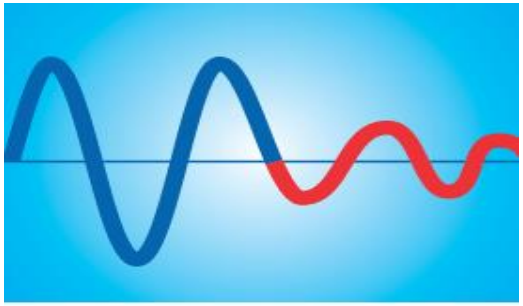


OVER VOLTAGE

- **Impacts:**
  - Electronic devices, burning of motor windings
  - Over heating of equipment
  - Bulbs, fuses blow off
  - Causes insulation failures resulting sparks in the circuit, etc.

Under Voltages

- Decrease in the rms ac voltage to less than 90% at the power frequency for a duration longer than 1 min.
- **Causes:**
  - Load switching ON
  - Capacitor switching OFF
  - System voltage regulation



BROWN OUT

- **Impacts:**
  - Motors may not start or run at low speed
  - Computer crashes
  - Filament bulbs will glow dim but fluorescent bulbs may not glow.
  - ma;loperation of refrigerators, etc.

Voltage Imbalance

- **Causes:**
  - Faulty operation of power factor correction equipment.
  - Unbalanced or unstable utility supply.
  - Unbalanced transformer bank supplying a three phase load that is too large for the bank.
  - Unevenly distributed single-phase loads on the three phase system.

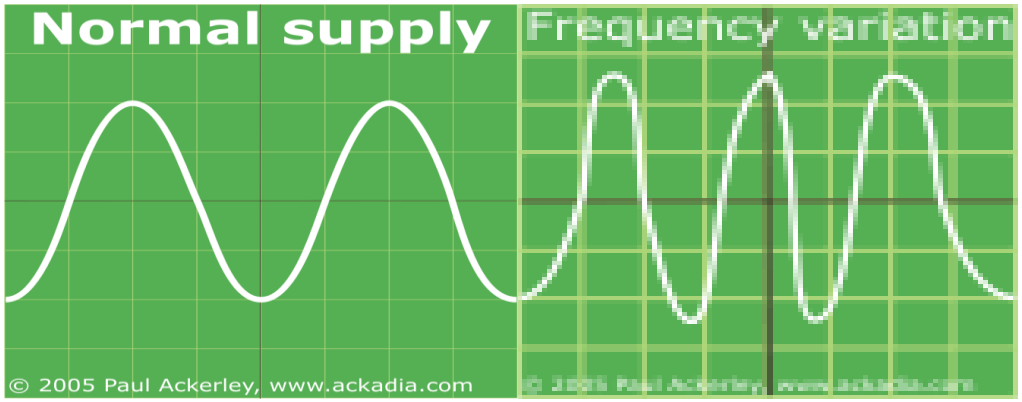
- **Effects:**
  - Degrades the performance
  - Causes phase current unbalance leading to torque pulsations
  - Increased vibrations, mechanical stresses and increased losses
  - Motor overheating which results in a shorter winding insulation life.

Interruptions

- **Causes:**
  - Temporary faults
  - Lightning strike
  - Tree limbs falling across conductors
- **Impacts:**
  - Operation interruption
  - Production losses
  - Revenue losses

Frequency Disturbances

- The deviation of the power system fundamental frequency from its specified nominal value (e.g. 50 or 60 Hz).
- **Causes:**
  - Poor speed regulations of local generation
  - Faults on the bulk power system
  - Large block of load being disconnected
  - Disconnecting a large source of generation



Importance of Maintaining Frequency

- Speed of motors changes according to frequency
- Affects the throughput of driven equipment
- Cogeneration units are affected since they have to follow mains system frequency
- Under/over frequency can prove harmful for the prime mover of generators.

Need for Power Quality Improvement

- Power quality problems can cause:
  - Equipment failures
  - Equipment life reduction
  - Mal-operation such as nuisance trip
  - Safety hazards (interruptions)
- Correct diagnosis is essential:
  - differentiating between problems due to power quality and other reasons
- Power quality problems have to be detected by the use of appropriate instruments.

Minimising Failures

- Improving the reliability of individual subsystems
- Installation of equipment as per relevant codes and standards
- Instituting periodic inspection and appropriate preventive/predictive maintenance regimes
- Maintaining records of work done, problems and failures
- Training of personnel to minimise human errors
- Minimising the impact of weather-related incidents

Minimise Duration of Interruption

- Providing measures such as auto reclosers
- Providing redundancy in design so that a single failure does not result in total disruption:
  - Redundant equipment (electrical and mechanical)
  - Redundant feeders/sections backed up by automation
  - Redundant sources backed up by automation
- Proving no-break take over systems (such as uninterrupted power supply equipment)

Handling Voltage Variations

- Equipment specifications matching with site characteristics :
  - design takes care of variations up to specified limits decided by actual observations in the supply system/determined by calculations
- Local improvements in specific parts of a plant:
  - Use of voltage regulating devices/UPS in specific circuits
- Total system improvements:
  - addition of generating capacity/tie feeders close to load bus

Mitigating Frequency Disturbances

- Separate generator for critical loads.
- Use of uninterrupted power system (UPS) whose frequency can be independently regulated.

End of Webinar 8

Any questions or comments?